When Covariates Are Censored: Methods Motivated by Huntington Disease

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Wake Forest University School of Medicine

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Department of Biostatistics, Johns Hopkins University
November 4, 2026

Slides available at thevazquezlab.com/talks

About me

Training

- BS Statistics and BA Economics, University of New Mexico (2019)
- MS (2023) and PhD (2025) Biostatistics, UNC Chapel Hill, with Tanya P. Garcia
- Provost's Postdoctoral Fellow, Biostatistics, Johns Hopkins (since 2025), with Elizabeth A. Stuart

Methodological and applied interests

- Methods for incomplete data: missing, censored, mismeasured, or spread across sites
- Applied biostatistics in neurodegenerative disease
- Collaborations in oncology, nephrology, pulmonary health, epilepsy imaging, and physical activity

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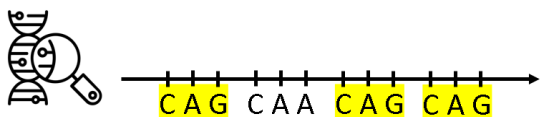
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Data I have worked with

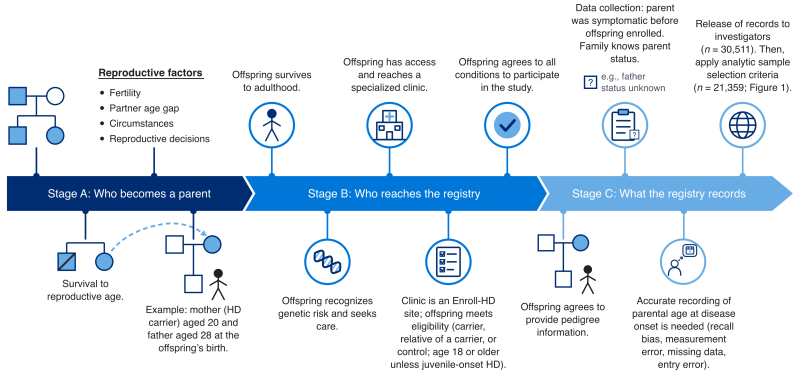
Huntington disease (HD)

- Inherited neurodegenerative disease caused by a CAG repeat expansion in the HTT gene; any individual with ≥ 40 CAG repeats will develop HD.
- Affects motor, cognitive, and psychiatric health. Clinical diagnosis is based on motor signs.
- Current treatments manage symptoms, but none delays or stops their onset.[1]



[1] Sarah J Tabrizi et al. "Potential disease-modifying therapies for Huntington's disease: lessons learned and future opportunities". In: *The Lancet Neurology* 21.7 (2022), pp. 645–658

Enroll-HD: who is in the data?



Participants reach Enroll-HD through HD clinics, so the cohort is a selected sample.

Jesus E. Vazquez et al. "Parental age at symptom onset and the sex of the affected parent in Enroll-HD: a hypothesis generating analysis." Working paper.

Cognition before an HD diagnosis

Research question

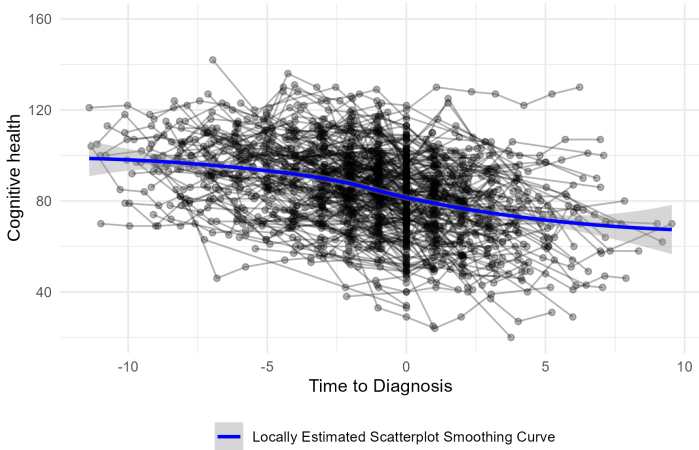
How does cognition change in the years before an HD diagnosis?

- Outcome Y: Stroop Colour Naming Test (SCNT) score in PREDICT-HD.
- Covariate X: age at diagnosis; time to diagnosis is $A - X$, with A the age at study entry.
- $n = 1{,}009$ participants; age at diagnosis is right-censored for 83%.

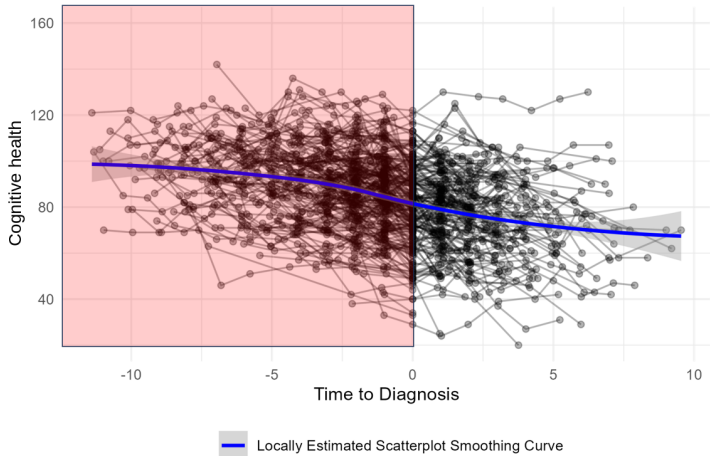
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RED GREEN
BLUE PURPLE

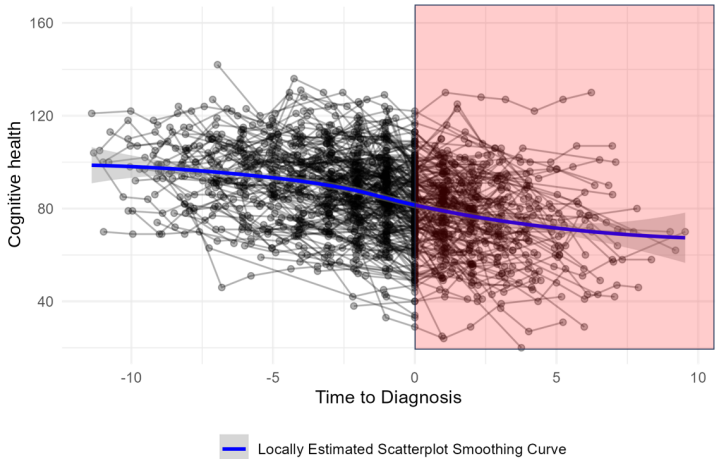
Why model decline before a diagnosis?



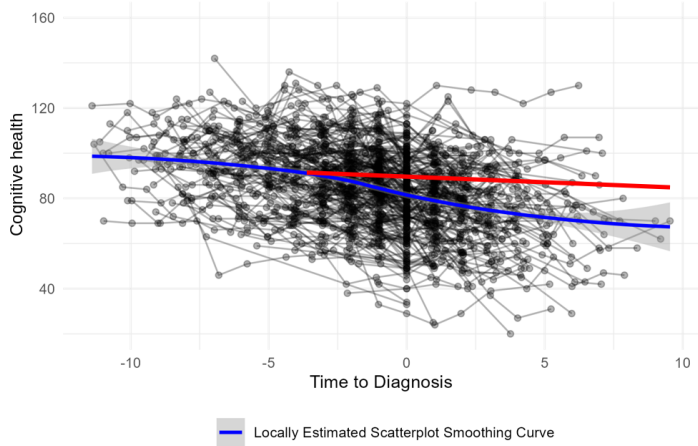
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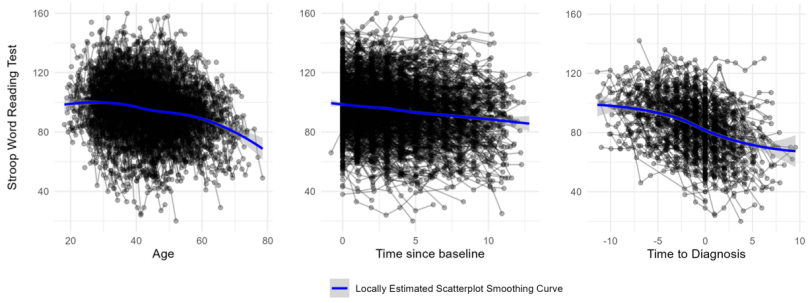
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Which time scale?



Each time scale answers a different question, and the shape of decline changes with the scale: anchored at diagnosis, the decline looks sigmoidal.

The problem appears across fields

Figure 1: Hormone levels by time to menarche[2]

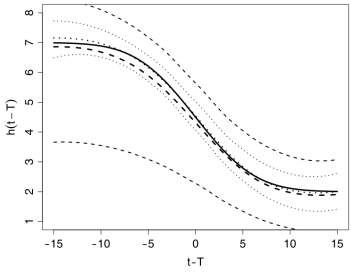
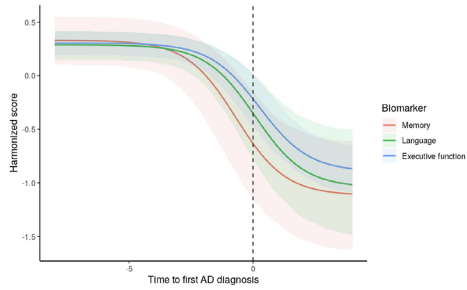


Figure 2: Cognitive markers by time to Alzheimer's diagnosis[3]



[2] Xuewen Lu et al. "Longitudinal data analysis with event time as a covariate". In: *Statistics in Biosciences* 2 (2010), pp. 65–80

[3] Kaidi Kang et al. "Comparing the dynamics of multiple cognitive markers for Alzheimer's disease progression using a novel double anchoring events-based sigmoidal mixed model". In: *Alzheimer's & Dementia* 20 (2024), e085501

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Figure 3: Educational attainment by parental age at HD onset[4]

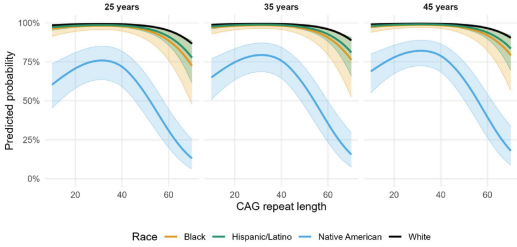
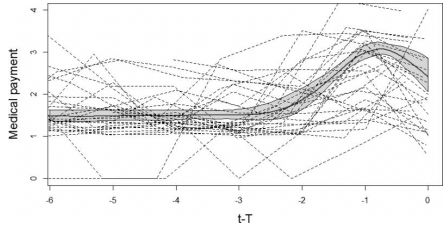


Figure 4: Medical payments by time to death[5]



[4] Jesus E Vazquez et al. "Higher educational attainment in Huntington disease families: evidence from the Enroll-HD study". In: *Orphanet Journal of Rare Diseases* 21.1 (2026), p. 142

[5] Shengchun Kong et al. "Conditional modeling of longitudinal data with terminal event". In: *Journal of the American Statistical Association* 113.521 (2018), pp. 357–368

How do we model an outcome indexed by time to an event that, for most people, has not happened yet?

A right-censored covariate

For subject i, let Y_i be the outcome (SCNT), X_i a covariate subject to right-censoring (age at diagnosis), and $\mathbf{Z}_i$ fully observed covariates:

$$Y_i = m(X_i, \mathbf{Z}_i; \boldsymbol{\beta}) + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma^2).$$

[6] Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

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Right-censoring

- Observe $W = \min(X, C)$ and $\Delta = I(X \leq C)$
- W carries partial information: $X > C$

Missingness

- Observe X or NA, with $R = I(X \text{ is known})$
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Goal and assumptions: *Estimate* $\boldsymbol{\theta} = (\boldsymbol{\beta}^T, \sigma)^T$ *from* $(Y_i, W_i, \Delta_i, \mathbf{Z}_i)$*, assuming* $Y \perp\!\!\!\perp C \mid (X, \mathbf{Z})$ *and* $X \perp\!\!\!\perp C \mid \mathbf{Z}$.[6]

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Why not just drop the censored participants?

The complete case (CC) estimator[7]

$$\sum_{i=1}^{n} \delta_i \mathbf{S}_{\boldsymbol{\theta}}^{F}(y_i, w_i, \mathbf{z}_i; \boldsymbol{\theta}) = \mathbf{0}, \quad \mathbf{S}_{\boldsymbol{\theta}}^{F} = \partial \log f_{Y|X,\mathbf{Z}} / \partial \boldsymbol{\theta}^{\mathrm{T}}$$

Consistent under our assumptions, but discards 83% of PREDICT-HD.

[7] Marissa C Ashner and Tanya P Garcia. "Understanding the Implications of a Complete Case Analysis for Regression Models with a Right-Censored Covariate". In: *The American Statistician* 78.3 (2024), pp. 335–344

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Weight each complete case by the inverse probability of observing X. Early simulations showed bias in some settings and none in others; the bias appeared only when the outcome Y entered the model for the weights.

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Question for the rest of the talk: *Which approaches for handling missing data carry over to a right-censored covariate, which need to change, and can we gain efficiency without losing robustness?*

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Borrowing from the missing data literature

IPW estimator[8]

$$\sum_{i=1}^{n} \frac{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^{F}(y_i, w_i, \mathbf{z}_i; \boldsymbol{\theta})}{\pi_{X,\mathbf{Z}}(w_i, \mathbf{z}_i)} = \mathbf{0}, \quad \pi_{X,\mathbf{Z}}(x, \mathbf{z}) = \mathsf{pr}(C > x \mid X = x, \mathbf{Z} = \mathbf{z})$$

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MLE[9]

$$\sum_{i=1}^{n} \Bigg[\underbrace{\delta_i \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, w_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Observed}} + \underbrace{(1-\delta_i) \frac{\int_{w_i < x} \mathbf{S}_{\boldsymbol{\theta}}^F(y_i, x, \mathbf{z}_i; \boldsymbol{\theta}) f_{Y|X,\mathbf{Z}}(y_i, x, \mathbf{z}_i; \boldsymbol{\theta}) f_{X|\mathbf{Z}}(x, \mathbf{z}_i)\, dx}{\int_{w_i < x} f_{Y|X,\mathbf{Z}}(y_i, x, \mathbf{z}_i; \boldsymbol{\theta}) f_{X|\mathbf{Z}}(x, \mathbf{z}_i)\, dx}}_{\text{Right-censored}} \Bigg] = \mathbf{0}$$

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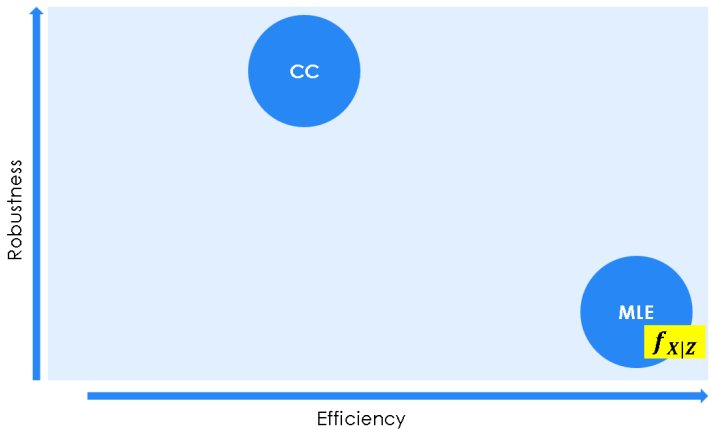
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Choosing between the two involves a trade-off between robustness and efficiency.

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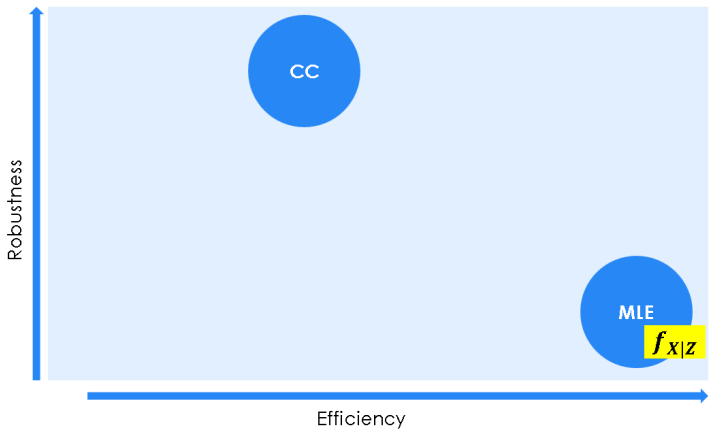
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Robustness vs. efficiency



Robustness vs. efficiency

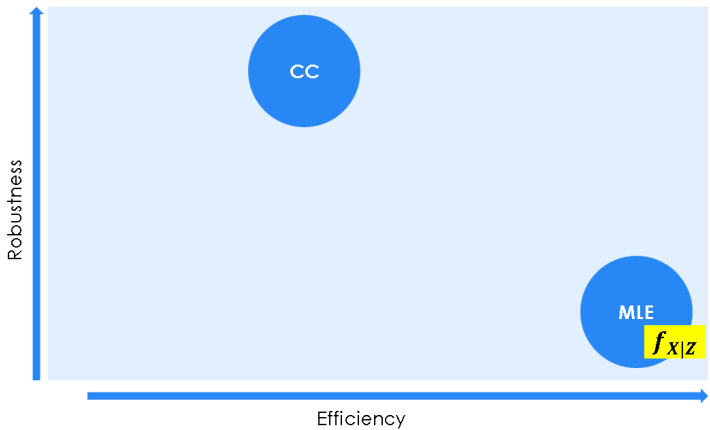
Where does the IPW estimator land?



Robustness vs. efficiency

Where does the IPW estimator land?

Can we find other estimators that are both robust and efficient?



Augmentation with guaranteed efficiency gains

Each estimator solves $\sum_{i=1}^{n} \mathbf{\Phi}_{\text{est}}(\mathbf{O}_i; \boldsymbol{\theta}) = \mathbf{0}$ and has influence function $\boldsymbol{\Upsilon}$. Add a mean-zero term to a consistent estimator to get $\boldsymbol{\Upsilon}_{\text{AUG}}$.[10]

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Efficiency condition

$$\mathbf{0} = \text{cov}(\boldsymbol{\Upsilon}_{\text{AUG}} - \boldsymbol{\Upsilon}, \boldsymbol{\Upsilon}_{\text{AUG}}) \quad \Longrightarrow \quad \text{var}(\boldsymbol{\Upsilon}) \geq \text{var}(\boldsymbol{\Upsilon}_{\text{AUG}}).$$

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Augmented complete case (ACC) estimator

Right-censored covariate[12]

$$\sum_{i=1}^{n}\Big[\underbrace{\delta_i \mathbf{S}^F_{\boldsymbol{\theta}}(y_i, w_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{CC}} + \underbrace{\{\delta_i - \pi_{Y,\mathbf{Z}}(y_i, \mathbf{z}_i)\}\, \boldsymbol{\Psi}^{\text{cens}}_{\text{ACC}}(y_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Augmentation}}\Big] = \mathbf{0}$$

Missing covariate[11]

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The augmentation terms differ[12]

$$\boldsymbol{\Psi}_{\text{ACC}}^{\text{cens}} = -\frac{E_{X|\,Y,\mathbf{Z}}\left\{\pi_{X,\mathbf{Z}}(X, \mathbf{z})\, \mathbf{S}_{\boldsymbol{\theta}}^{F}(y, X, \mathbf{z}; \boldsymbol{\theta})\right\}}{\pi_{Y,\mathbf{Z}}(y, \mathbf{z})}$$

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Two more augmented estimators

Modified ACC (MACC): augments the CC estimator[12]

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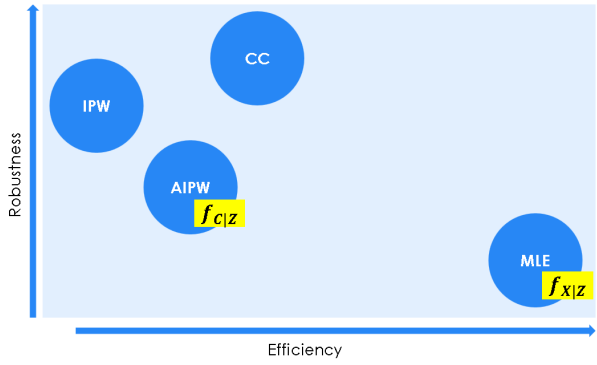
Augmented IPW (AIPW): augments the IPW estimator[13]

$$\sum_{i=1}^{n}\Big[\underbrace{\frac{\delta_i \mathbf{S}^F_{\boldsymbol{\theta}}(y_i, w_i, \mathbf{z}_i; \boldsymbol{\theta})}{\pi_{X,\mathbf{Z}}(w_i, \mathbf{z}_i)}}_{\text{IPW}} + \underbrace{\{1-\delta_i/\,\pi_{X,\mathbf{Z}}(w_i, \mathbf{z}_i)\}\boldsymbol{\Psi}^{\text{cens}}_{\text{AIPW}}(y_i, \mathbf{z}_i; \boldsymbol{\theta})}_{\text{Augmentation}}\Big] = \mathbf{0}$$

12 Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

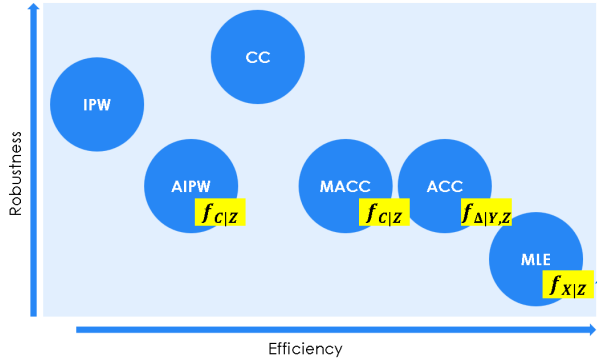
13 J.M. Robins, A. Rotnitzky, and L.P. Zhao. "Estimation of regression coefficients when some regressors are not always observed". In: *Journal of the American Statistical Association* 89.427 (1994), pp. 846–866

Robustness vs. efficiency[12]



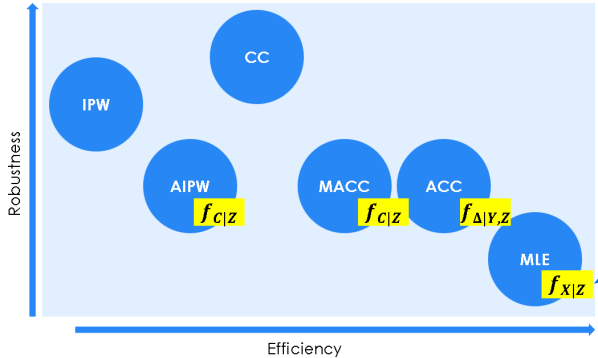
12 Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

Robustness vs. efficiency[12]



[12] Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

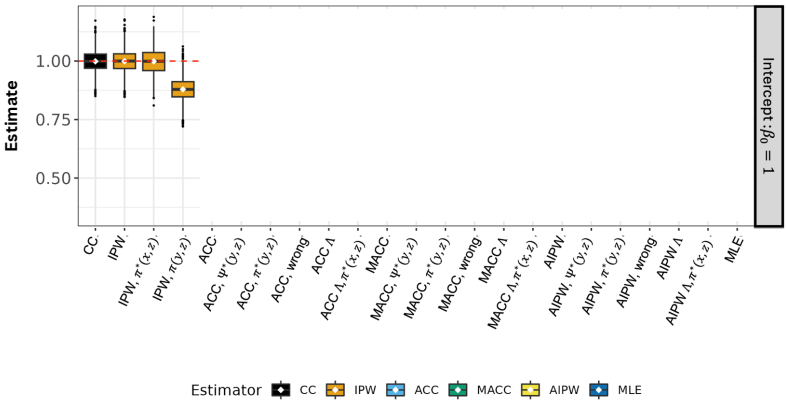
Robustness vs. efficiency[12]



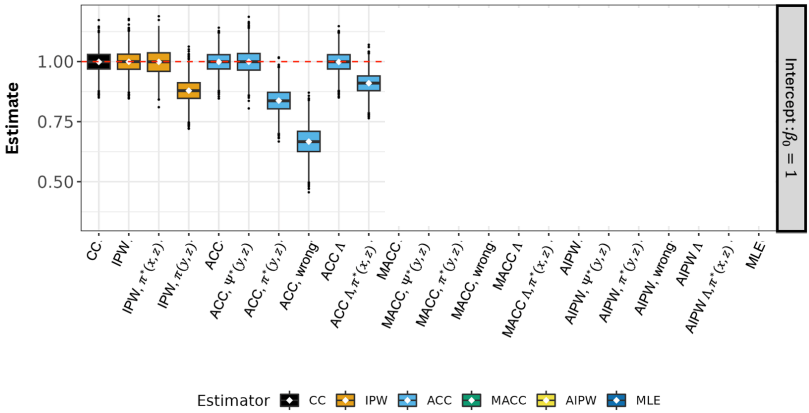
Variance ranking: $\text{MLE} < \begin{matrix}\text{ACC} \\ \text{MACC}\end{matrix} < \text{CC} \leq \text{AIPW} < \text{IPW}$

[12] Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

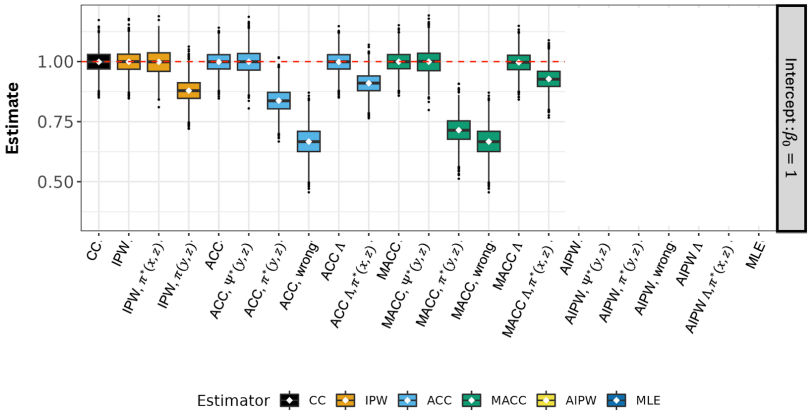
Simulation results: bias



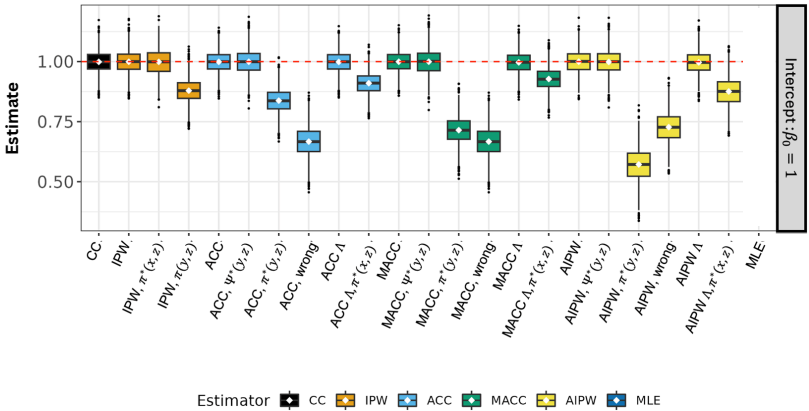
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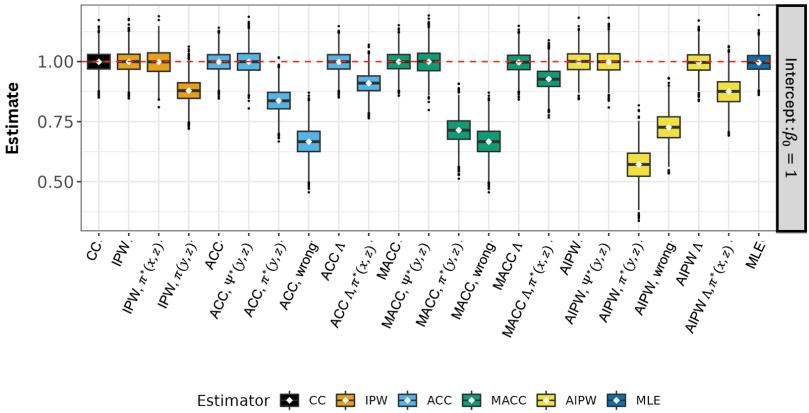
Simulation results: bias



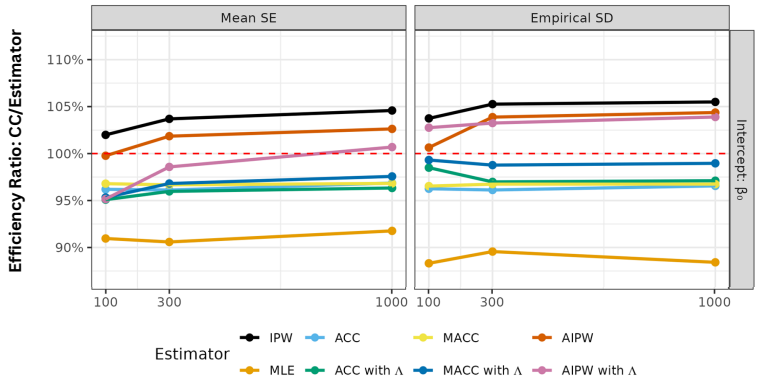
Simulation results: bias



Simulation results: bias



Simulation results: efficiency



Back to the data: PREDICT-HD

$$\text{SCNT} = \beta_0 + \beta_1(A - X) + \beta_2 Z_{\text{CAG}} + \beta_3 Z_{\text{Educ}} + \beta_4 Z_{\text{Female}} + \beta_5 Z_{\text{Eye}} + \epsilon$$

Estimator	Time to diagnosis	CAG	Education	Sex (female)	Eye movement
CC	−1.20 (0.29)**	−0.90 (0.27)**	0.84 (0.32)**	−1.24 (1.83)	−2.64 (1.63)*
IPW	−1.16 (0.32)**	−0.75 (0.31)*	0.82 (0.38)*	−2.25 (2.27)	−3.03 (1.89)*
MACC	−1.07 (0.24)**	−1.21 (0.28)**	0.77 (0.29)**	−1.50 (1.55)	−4.63 (1.45)**
AIPW	−1.16 (0.22)**	−0.75 (0.12)**	0.83 (0.26)**	−0.86 (1.38)	−3.32 (1.43)*
ACC	−1.19 (0.28)**	−0.90 (0.22)**	0.84 (0.22)**	−1.15 (1.22)	−2.64 (1.04)*
MLE	−1.20 (0.10)**	−0.91 (0.22)**	0.85 (0.21)**	−1.24 (1.22)	−2.65 (1.13)*

Stroop Color Naming Test (SCNT), PREDICT-HD (n = 1,009). Significant at the 0.01 (**) or 0.05 (*) level.

Back to the data: PREDICT-HD

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- All estimators agree: cognition declines as participants approach diagnosis.

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Stroop Color Naming Test (SCNT), PREDICT-HD (n = 1,009). Significant at the 0.01 (**) or 0.05 (*) level.

- All estimators agree: cognition declines as participants approach diagnosis.
- Standard errors follow the variance ranking: the MLE is smallest and the IPW estimator is largest.

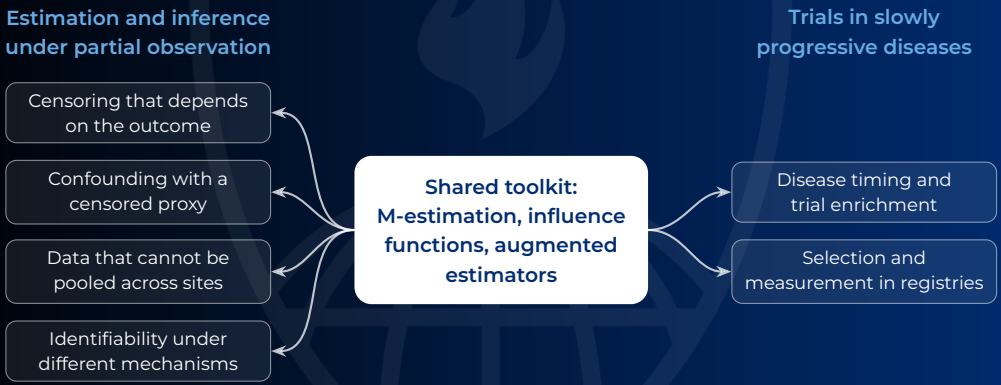
Recommendations

The choice depends on the tolerance for misspecified nuisance distributions. From most to least preferred:

1. **The ACC estimator:** more efficient than the CC and IPW estimators; needs only $f_{\Delta|Y,\mathbf{Z}}$, and $(Y, \Delta, \mathbf{Z})$ are always observed.
2. **The MACC estimator:** more efficient than the CC estimator; best when independent censoring holds and $f_{C|\mathbf{Z}}$ is well specified.
3. **The CC estimator:** the most robust; include it in every analysis as a diagnostic.
4. **The MLE:** the most efficient when $f_{X|\mathbf{Z}}$ is correct, but biased when misspecified.
5. **The IPW and AIPW estimators:** less efficient than the CC estimator, with no gain for this problem.

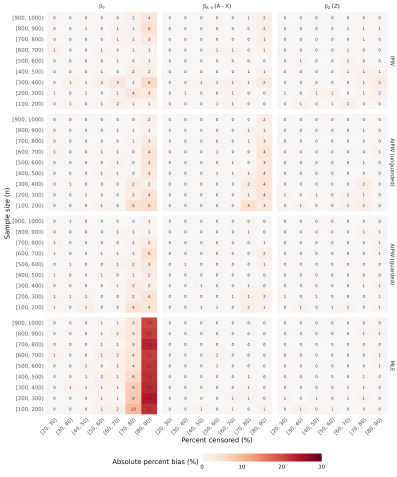
LOOKING AHEAD

One toolkit, two research directions



Long-term goal: a methods program for rare-disease and multi-site clinical research.

Next: outcome-dependent censoring[14]



Healthier participants are diagnosed later, so censoring depends on the outcome.

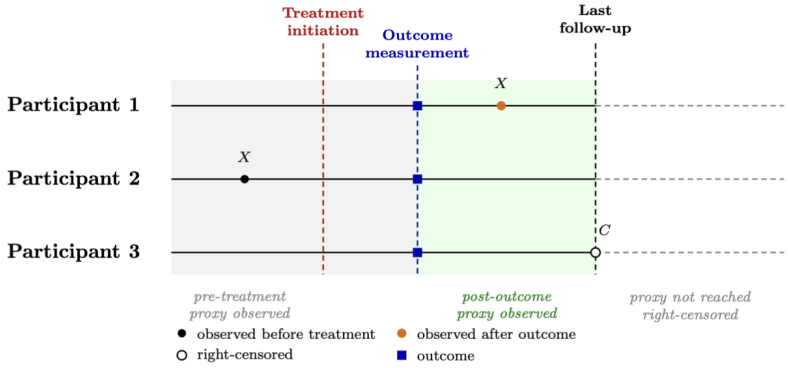
Assumptions:

- $C \not\perp\!\!\!\perp Y \mid (X, \mathbf{Z})$: censoring may depend on the outcome.
- $C \perp\!\!\!\perp X \mid (Y, \mathbf{Z})$: given Y and $\mathbf{Z}$, censoring is unrelated to the age at diagnosis.

Under review at Statistics in Medicine.

14 Jesus E Vazquez et al. "Robust Estimation under Outcome Dependent Right Censoring in Huntington Disease: Estimators for Low and High Censoring Rates". In: *arXiv preprint arXiv:2511.15929* (2025)

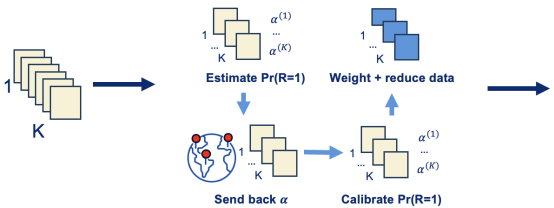
Next: censored proxies in causal inference



A disease-progression proxy X that may help adjust for confounding is often right-censored. *(2026+)*

Jesus E. Vazquez et al. "Causal inference with a right-censored marker of disease progression: adjusting for latent severity in Huntington disease." Working paper.

Next: data spread across sites[15]



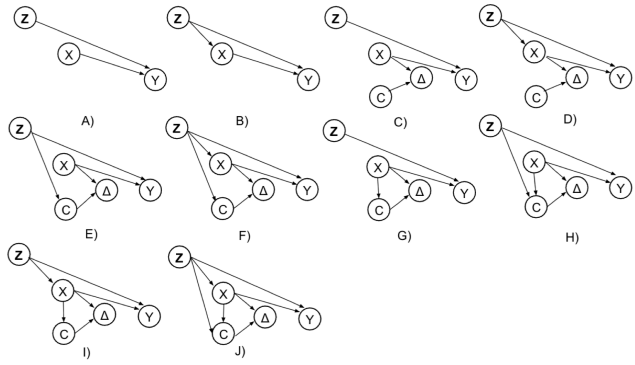
$$\beta_{pooled} = \left[\sum_{k=1}^{K}\{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\}^{\top}\{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\}\right]^{-1}\left[\sum_{k=1}^{K}\{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\}^{\top}Y^{(k)}\right]$$

Federated estimation when values are missing and sites cannot pool data.

Under review at Statistics in Medicine.

[15] Jesus E Vazquez et al. "Federated Learning with Incomplete Data: When to Use Complete Cases and When to Weight". In: *arXiv preprint arXiv:2605.20125* (2026)

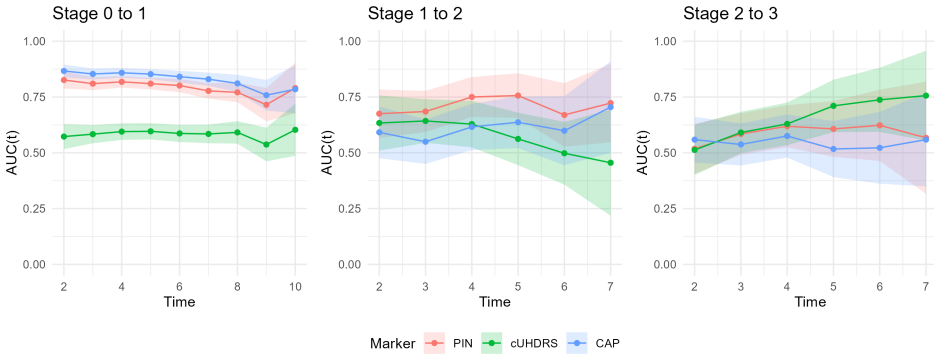
Next: identifiability with DAGs



Which nuisance distributions are identifiable under each censoring mechanism? With Dewei Lin. *(2026+)*

Dewei Lin et al. "Identifiability of regression models with a right-censored covariate: extending canonical causal diagrams." Working paper.

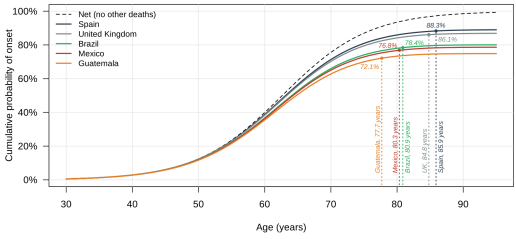
Next: disease timing and trial enrichment

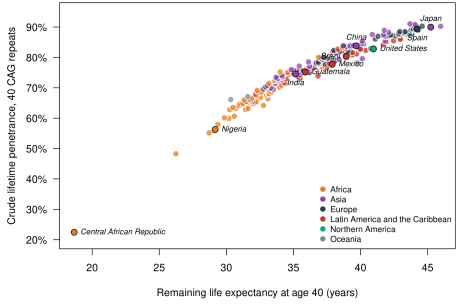


Which marker best enriches a trial at each disease stage? With Madhuri Raman. *(2026+)*

Madhuri Raman et al. "Predicting Huntington disease stage transitions for clinical trial enrichment." Working paper.

Applied work: HD penetrance around the world

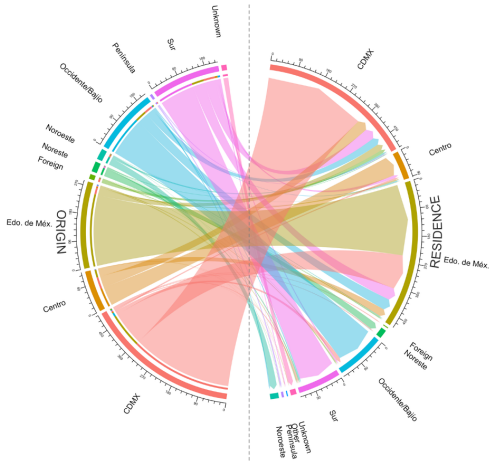




How penetrant is HD outside European-ancestry cohorts? *(2026+)*

Jesus E. Vazquez et al. "Fully penetrant within whose lifetime? Background mortality and the penetrance of Huntington's disease alleles." Working paper.

Applied work: HD in Mexico



Characterizing HD in Mexico with clinicians in Mexico City. *(2026+)*

I lead the statistical work:

- Patients with more resources tend to live closer to the referral center and are more likely to be seen.
- The clinic cohort may not represent the wider HD community, which matters for transportability.

Adriana Ochoa-Morales, Jesus E. Vazquez, et al. "Clinical, molecular, and survival characteristics of Huntington's disease in Mexico: a 30-year cohort from the National Referral Center." Working paper.

Service to the field



Planning Committee, Fostering Excellence in Biostatistics Workshop (ENAR 2026, Indianapolis)



Co-founder, Latine Graduate and Professional Student Association at UNC Chapel Hill (now over 400 members)

The International Biometric Society ENAR Social Media Chair (2024–2025)

Mentor, Científico Latino (2026)

Thank you!

Mentors

Tanya P. Garcia (PhD advisor) and Elizabeth A. Stuart (postdoctoral advisor).

Collaborators

Marissa Ashner, Yanyuan Ma, Karen Marder, Sarah Lotspeich, Brian Richardson, Madhuri Raman, Dewei Lin, Jiayi Tong, Yicheng Shen, and the Methods for Incomplete Data Lab.

Funding

National Institute of Environmental Health Sciences: **P30ES010126** (TPG), **T32ES007018** (JEV); National Institute of Neurological Disorders and Stroke: **R01NS131225** (TPG, JEV); Johns Hopkins Provost's Postdoctoral Fellowship (JEV).

Questions?

Slides available at thevazquezlab.com/talks

Supplementary Material

Simulation set-up

$$Y_i = \beta_0 + \beta_1(A_i - X_i) + \beta_2 Z_i + \epsilon_i, \quad (X_i, C_i, Z_i)^T \sim \text{Normal}\left\{ \mathbf{0}, \begin{bmatrix} 1 & 0.6 & 0.3 \\ 0.6 & 1 & 0.5 \\ 0.3 & 0.5 & 1 \end{bmatrix} \right\},$$

with $A_i \sim \text{Normal}(0, 1)$, $\epsilon_i \sim \text{Normal}(0, 1)$, so that $\rho_{X,C|Z} = 0$.

- $\boldsymbol{\theta} = (\beta_0, \beta_1, \beta_2, \sigma)^T = (1, 3, 2, 1)^T$
- $n = 1{,}000$ and 3,000 replicates
- Nuisance densities correctly specified or misspecified in turn

What each estimator needs, and what it buys[12]

	Right-censored covariate problem: $f_{X,C,Z}$						Missing covariate problem: $f_{X,R,Z}$						
Estimator	Independent		Dependent				Independent		Dependent				Other
	$f_{X\|Z}$	$f_{C\|Z}$	$f_{X\|C,Z}$	$f_{C\|Z}$	$f_{C\|X,Z}$	$f_{X\|Z}$	$f_{X\|Z}$	$f_{R\|Z}$	$f_{X\|R,Z}$	$f_{R\|Z}$	$f_{R\|X,Z}$	$f_{X\|Z}$	f_Z
CC	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IPW	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
MLE	×	✓	×	✓	×	×	×	✓	×	✓	×	×	✓
ACC	$f_{\Delta=1\|Y,Z}$ must be correct		$f_{\Delta=1\|Y,Z}$ must be correct				One must be correct		$f_{R=1\|Y,Z}$ must be correct				✓
MACC	✓	×	×	×	×	✓	One must be correct		×	×	×	✓	✓
AIPW	✓	×	×	×	×	✓	One must be correct		×	×	×	✓	✓
			Option 1		Option 2				Option 1		Option 2		

✓: the density can be misspecified; ×: the density must be correctly specified.

12 Jesus E Vazquez et al. "How Estimators Change When Adapted From the Missing Covariate Problem to the Right-Censored Covariate Problem". In: *International Statistical Review* (2026)

Most participants are not yet diagnosed

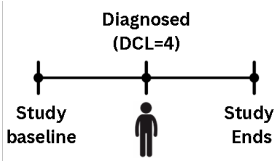


Figure 5: Age at diagnosis is observed

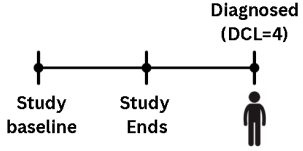


Figure 6: Age at diagnosis is right-censored

Age at diagnosis X is right-censored for 83% of PREDICT-HD participants.

Consistency and asymptotic normality

Theorem (1)

Assume $(C, R) \perp\!\!\!\perp Y | X, \mathbf{Z}$. *Let* $\widehat{\boldsymbol{\theta}}$ *solve* $\sum_{i=1}^{n} \boldsymbol{\Phi}(\mathbf{O}_i; \boldsymbol{\theta}) = \mathbf{0}$ *for one of the six estimators. Under standard regularity conditions,*

$$\sqrt{n}(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \rightarrow \textit{Normal}(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-\mathrm{T}}),$$

where $\mathbf{A} = E\{\partial \boldsymbol{\Phi}(\mathbf{O}; \boldsymbol{\theta}_0)/\partial \boldsymbol{\theta}^{\mathrm{T}}\}$ *and* $\mathbf{B} = E\{\boldsymbol{\Phi}(\mathbf{O}; \boldsymbol{\theta}_0)^{\otimes 2}\}$, *with influence function* $\boldsymbol{\Upsilon} = -\mathbf{A}^{-1}\boldsymbol{\Phi}(\mathbf{O}; \boldsymbol{\theta}_0)$.